

It Only Takes *OneBite*: Attention Shocks in Local Markets^{*}

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Abstract:

In an era of information abundance, scarce attention constrains consumer choice. Exploiting a unique quasi-experimental setting, I evaluate how viral attention from an online *OneBite* pizza review affects consumer decisions and the mechanisms that explain the consumer response. Using a staggered event study, I estimate that total customers at the reviewed pizzeria increase 6.2% in the first month following the review. However, the increase critically depends on the review score. While scores < 7.0 and scores in the range $[7.0, 8.0)$ do not increase monthly customers, scores ≥ 8.0 increase customers by an average of 43% in each of the first three months following the review. The surge in customer traffic generates positive spatial spillovers onto nearby drinking places, consistent with consumer-based agglomeration from the shared customer base. Despite the initial increase, viral attention fades as monthly customers return to their pretreatment level within four months. Using a bounded consideration model, I separate the demand response into an attention channel that increases awareness of the pizzeria and an information channel that updates a consumer's belief about its quality. Simulated maximum likelihood estimates indicate that although the score changes perceived quality, increased attention is the dominant mechanism to explain the increase in customers.

Keywords: Attention Economy, Limited Consideration, Social Media, Spatial Spillovers
JEL Codes: D12, D8, M3, R10

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1 Introduction

Information abundance defines modern consumer markets, with a seemingly infinite supply of information about any given product. Although the wealth of information creates an informed consumer, scarce attention limits how much information a consumer can process and restricts which products consumers consider (Simon, 1971). In this attention economy, the media’s persuasive power to draw attention to products can fundamentally shape consumer behaviour and market outcomes. However, understanding how media attention versus product information influences consumer choice has received limited attention.

In this paper, I exploit a unique quasi-experimental setting to investigate how an increase in attention on social media from a *OneBite* pizza review affects consumer decisions. In each *OneBite* review, Barstool Sports’ Dave Portnoy reviews a cheese pizza at a local pizzeria, scores the pizzeria from 0.0 to 10.0, and uploads the video review to YouTube. Since the pizzeria cannot anticipate the online review, the attention from the viral video review provides a quasi-exogenous firm-level demand shock. To evaluate how the attention shocks affect consumer demand, I create a comprehensive dataset of all *OneBite* review scores and video dates from both the *OneBite* app and YouTube, and link the reviews to firm-level outcomes in the United States from SafeGraph’s Spend Patterns dataset. Using a staggered event study framework, I estimate that total customers increase by 6.2% in the first month after the review. However, the attention shock dissipates, with no detectable customer increase in later months. I next evaluate heterogeneous treatment effects across three score categories based on Portnoy’s score guide (<7.0 , $[7.0, 8.0)$, and ≥ 8.0) and determine that the effects critically depend on the review score. Although total customers do not increase after a pizzeria receives a score <7.0 or in $[7.0, 8.0)$, review scores ≥ 8.0 increase customers by a substantial +48%, +47%, and +34% in the first three months after the review. Despite the significant increase, monthly customers return to their pretreatment level in the fourth month after the review. While attention on social media generates short-term benefits for firms, no long-term benefits exist as viral attention fades.

I next evaluate the broader local-market effects of a *OneBite* review and investigate whether the attention spills over to nearby establishments, following Butts (2021). I first investigate whether consumers substitute away from local competitors and jointly estimate the direct treatment effect on the reviewed pizzeria, alongside the spillover effects onto pizzerias in two treatment rings: within 500m, and between 500m and 1,000m of the reviewed pizzeria. In the event study with spatial spillovers, the direct treatment ef-

fect remains statistically similar to the main estimate, while nearby pizzerias experience no detectable spillover effects. Furthermore, I do not detect any spillovers onto nearby competitors following a review score ≥ 8.0 . The unchanged customer levels at nearby untreated pizzerias indicate that the increase in customers from a *OneBite* review is consistent with market expansion, rather than business stealing, as it draws new consumers into the market.

With the influx of customers to the local market, I investigate whether a *OneBite* review generates positive spillovers for nearby establishments in other industries. Over the first three months following the review, customers increase marginally at “Snack and Non-Alcoholic Beverage Bars” (+6%) and “Beer, Wine, and Liquor Stores” (+2%), but decrease at “Limited-Service Restaurants” (−1%). However, high-score reviews generate more significant spillovers, as customers increase at nearby “Drinking Places (Alcoholic Beverages)” (+7%) and decrease at “Limited-Service Restaurants” (−7%). The spillovers suggest that while customers do not substitute away from other pizzerias, customers substitute away from limited-service restaurants that serve other dining options. Furthermore, the increase in customers at “Drinking Places (Alcoholic Beverages)” is consistent with consumption-based agglomeration because the positive demand shock to an establishment expands the local market and generates spillovers onto other nearby establishments. As the increase in drinking places likely originates from Barstool Sports social media followers, establishments that provide complimentary products for the new patrons will benefit from the demand shock. Since Barstool Sports often produces media content on sports and topics people would “talk about while sitting at a bar,” a *OneBite* review’s audience likely overlaps with patrons likely to visit drinking establishments. The positive spillover suggests the viral attention both generates benefits for the treated pizzeria and spurs economic activity for complementary establishments in the local area.

Although the increase in customers shows that viral attention affects consumer decisions, the effects vary based on a consumer’s prior familiarity with the market, as prior knowledge of local alternatives may weaken the effects of attention (Ackerberg, 2001). To investigate how market familiarity affects the consumer response, I evaluate how the treatment effects vary for non-local consumers with knowledge about the alternatives in the market. Although high-score reviews generate increases in customers from all distance bins, the treatment effect is monotonic in distance, with the largest increase from non-local customers (+65%) who live more than 50km from the pizzeria’s county. The greater response from non-locals provides suggestive evidence that attention increases awareness of the reviewed pizzeria in the market.

Finally, I supplement the reduced-form analysis with a bounded consideration model that adapts [Goeree \(2008\)](#) to separate the attention and information channels from a *OneBite* review. While viral attention changes the probability that a consumer considers the pizzeria, the differences between the review score and the pizzeria’s pre-review Google rating provide information for consumers to update their expected utility. Using a simulated log-likelihood, I determine that while the change in perceived quality only marginally shifts a consumer’s conditional choice probabilities, the viral attention from high-score reviews creates a significant increase in consumer consideration. Similarly, low-score reviews experience an increase in consideration; however, negative information offsets any potential increase. Although reviews both generate attention and convey product information, attention governs consumer choice.

This paper makes three main contributions. First, I provide the first causal firm-level evaluation to show how attention shocks affect local business performance and decompose the effects into multiple mechanisms. In the attention economy, understanding how viral attention affects firm performance remains a foundational question in the literature as firms compete for scarce consumer attention [Loewenstein and Wojtowicz \(2025\)](#); [Bordalo et al. \(2016\)](#); [Simon \(1971\)](#). While the viral attention shocks increase consumer awareness about a product ([Helmers et al., 2019](#); [Honka et al., 2017](#); [Goeree, 2008](#)), *OneBite* review scores themselves provide information to consumers about product quality and introduce a secondary channel that affects consumer decisions. Related literature on advertising shows that information changes a consumer’s perception about product quality ([Dubé et al., 2026](#); [Bagwell, 2007](#)), and outlines the key distinction between persuasion and information ([Akerberg, 2003, 2001](#)). My paper separates how attention and information contribute to consumer demand and shows that the increase in product consideration drives the consumer response.

Second, I bridge the gap between online review platform literature ([Bao et al., 2024](#); [Fang, 2022](#); [Lawani et al., 2019](#); [Kuang, 2017](#); [Hu et al., 2008](#)) and the consumption-based amenities literature ([Couture and Handbury, 2020](#); [Couture, 2016](#); [Glaeser et al., 2001](#)). While the agglomeration literature often focuses on the productivity benefits from clustering, consumption-based agglomeration benefits stem from a shared customer base in a market because the positive demand shock to an amenity expands the local market and increases firm performance at nearby establishments with complementary products. The positive spatial spillovers from online reviews I estimate provide direct causal evidence on the consumption-agglomeration mechanism in the literature.

Third, I provide new evidence on how media influences consumer decisions. Previous literature shows that radio (Wang, 2021) and television (Cornelson, 2018; Kearney and Levine, 2015; Ferrara et al., 2012) shape consumer behaviour through information. With social media's surge as an information hub (Levy, 2021; Allcott and Gentzkow, 2017), platforms offer a new space for influencers to shape consumption habits through subjective opinions, as audiences often interpret media opinions as facts (Bursztyn et al., 2023). Relative to the previous literature on other media, I estimate how viral social media posts influence consumer decisions through quasi-exogenous firm-level demand shocks. While attention from social media personalities directly relates to celebrity endorsements (Knoll and Matthes, 2017; Garthwaite, 2014; Elberse and Verleun, 2012; Agrawal and Kamakura, 1995), my unique setting provides additional insights into an endorsement's intensive margin from the review score. Furthermore, since Portnoy is not compensated for the review, the treatment effects capture the maximum willingness to pay for a celebrity or influencer endorsement.

The remainder of the paper proceeds as follows. Section 3 provides an overview of *OneBite* reviews. Section 4 describes the data and sample. Section 5 outlines the empirical strategy and identification. Section 6 presents the main results and robustness checks. Section 7 develops a bounded consideration model to investigate the mechanisms that explain the treatment effect. Finally, Section 8 concludes.